Lesley (Yajie) Zhou

Ph.D. Candidate in Computer Science, University of Maryland, College Park
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Research Interests

AI Reliability for Systems; Autonomous Network and System Operations; Robust AI-Driven Systems.

Education

University of Maryland, College Park 2023 – Present
Ph.D. in Computer Science
Advisor: Prof. Zaoxing (Alan) Liu

Boston University 2021 – 2023
Ph.D. in Computer Engineering
Advisor: Prof. Zaoxing (Alan) Liu

Korea Advanced Institute of Science and Technology (KAIST) 2020
M.S. in Electrical Engineering
Advisors: Prof. Yung Yi and Prof. Dongsu Han

Xidian University 2018
B.E. in Computer Science and Technology

Research Experience

Carnegie Mellon University Oct 2025 – Present
Visiting Researcher
Host: Prof. Vyas Sekar

Microsoft Research, Redmond, WA Summer 2025
Research Intern, Network Research Group
Mentors: Siva Kesava Reddy Kakarla, Ryan Beckett, Behnaz Arzani

Microsoft Research, Redmond, WA Summers 2023 & 2024
Research Intern, Network Research Group
Mentors: Kevin Hsieh, Sathiya Kumaran Mani, Srikanth Kandula

University of Chicago, Remote 2020 – 2021
Research Intern
Mentors: Prof. Junchen Jiang, Prof. Francis Y. Yan

Publications

Conference and Workshop Papers (Ordered by date)

[C1] Yen-Ru Chin, Milind Srivastava, **Yajie Zhou**, Giulia Fanti, Vyas Sekar. *T-pack: Structure-aware Generative Compression for Efficient Distributed Tracing*. USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2027.

[C2] **Yajie Zhou**, Jiajun Ruan, Eric S. Wang, Sadjad Fouladi, Francis Y. Yan, Kevin Hsieh, Zaoxing Liu. *NetArena: Dynamic Benchmarks for AI Agents in Network Automation*. International Conference on Learning Representations (ICLR), 2026.

[C3] **Yajie Zhou**, Fuheng Zhao, Eric S. Wang, Ayse Coskun, Divyakant Agrawal, Amr El Abbadi, Zaoxing Liu. *PrvTel: Lightweight Models for Private and Accurate Telemetry Data Retention*. USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2026.

[C4] **Yajie Zhou**, Nengneng Yu, Tuo Zhao, Zaoxing Liu. *Tidal: Tackling Concept Drift in Provenance-based Advanced Persistent Threats Detection*. New Ideas in Networked Systems (NINeS), 2026.

[C5] **Yajie Zhou**, Kevin Hsieh, Sathiya Kumaran Mani, Srikanth Kandula, Zaoxing Liu. *MeshAgent: Enabling Reliable Network Management with Large Language Models*. ACM SIGMETRICS International Conference on Measurement and Modeling of Computer Systems, 2026.

[C6] Sathiya Kumaran Mani, Kevin Hsieh, Santiago Segarra, Ranveer Chandra, **Yajie Zhou**, Srikanth Kandula. *Securing Public Cloud Networks with Efficient Role-based Micro-Segmentation*. USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2025.

[C7] **Yajie Zhou***, Nengneng Yu*, Zaoxing Liu. *Towards Interactive Research Agents for Internet Incident Investigation*. ACM Workshop on Hot Topics in Networks (HotNets), 2023.

[C8] Sathiya Kumaran Mani, **Yajie Zhou**, Kevin Hsieh, Santiago Segarra, Trevor Eberl, Eliran Azulai, Ido Frizler, Ranveer Chandra, Srikanth Kandula. *Enhancing Network Management Using Code Generated by Large Language Models*. ACM Workshop on Hot Topics in Networks (HotNets), 2023.

[C9] **Yajie Zhou***, Zhengxu Xia*, Francis Y. Yan, Junchen Jiang. *Genet: Automatic Curriculum Generation for Learning Adaptation in Networking*. ACM SIGCOMM Conference, 2022.

* denotes equal contribution.

Demos, Posters and Short Papers

[S1] Fenghao Dong, Yucheng Yin, **Yajie Zhou**, Peilin Xin, Shinan Liu, Vyas Sekar. *DeePCAP: Enabling High-Fidelity and Cost-Efficient Archival Packet Trace Storage*. ACM SIGCOMM Conference (Poster), 2026.

[S2] **Yajie Zhou**, Francis Y. Yan, Kevin Hsieh, Zaoxing Liu. *NetArena Adaptation for Next Waves of Network Benchmarks*. ACM SIGCOMM Conference (Demo), 2026.

[S3] **Yajie Zhou**, Jason Li, Gianluca Stringhini, Ayse K. Coskun, Zaoxing Liu. *Enabling Privacy-preserving Multidimensional Network Telemetry with Autoencoders*. IEEE International Conference on Cloud Engineering (IC2E), 2023.

Preprints

[R1] **Yajie Zhou**, Ao Li, Ashwin Silla, Zaoxing Liu, Vyas Sekar. *AIChilles: Automatically Uncovering Hidden Weaknesses in AI-Evolved Systems*. arXiv:2606.15834, 2026.

[R2] Milind Srivastava, Zeying Zhu, **Yajie Zhou**, Yancheng Yuan, Fenghao Dong, Peilin Xin, Zaoxing Liu, Vyas Sekar. *ASAP: Reimagining the Data Lifecycle using Application Semantic-Aware Processing*. arXiv:2608.12735, 2026.

[R3] Jialin Zhang, Fenghao Dong, **Yajie Zhou**, Vyas Sekar, Shinan Liu. *TabQueryBench: A Query-Centric Benchmark for Synthetic Tabular Data*. arXiv:2607.03926, 2026.

[R4] Sayan Sinha, Vipul Harsh, **Yajie Zhou**, Marko Morrison, B. Aditya Prakash, Vyas Sekar, Hui Zhang. *Constrained Creativity for SysOps Agents*. Preprint, Carnegie Mellon University, 2026.

Patents

[P1] Sathiya Kumaran Mani, **Yajie Zhou**, Kevin Hsieh, Santiago Segarra, Ranveer Chandra, Srikanth Kandula. *Graph Analysis and Manipulation*. US Patent Application MS#413451-US-NP, Published January 2025.

[P2] **Yajie Zhou**, Kasim Te. *Method and Apparatus for Transmitting Video Data*. Korean Patent Application 10-2020-0141018, Published October 2020.

HONORS AND AWARDS

Ann G. Wylie Dissertation Fellowship, University of Maryland (top 2 in CS)	2026
Winner, Berkeley AgentX–AgentBeats Competition, Coding Agent Track (NetArena)	2026
KAUST Rising Stars in AI Symposium (top 6% of 400+ global applicants)	2026
N2Women Young Researcher Fellowship	2022, 2026
Graduate School Research Fellowship at KAIST	2018-2020
Outstanding Bachelor Dissertation Award, Xidian University (top 1% in CS)	2018

INVITED TALKS

Carnegie Bosch Institute (host: Dr. Jorge Guajardo Merchan)	Dec 2026
Carnegie Mellon University (host: Parallel Data Lab)	Nov 2026
International Telecommunication Union (ITU) Workshop	Oct 2026
Princeton University (host: Prof. Maria Apostolaki)	Oct 2026
University of Illinois Urbana-Champaign (host: Prof. Francis Y. Yan)	Oct 2026
University of Michigan, Ann Arbor (host: Prof. Ang Chen)	Jun 2026
The University of Hong Kong (host: Prof. Shinan Liu)	Feb 2026

MEDIA COVERAGE

AI-Driven Research for Systems, UC Berkeley Sky Computing Lab (featuring AIChilles)	Jul 2026
Testing AI Before It Touches the Network, UMD Department of Computer Science	Mar 2026

TEACHING EXPERIENCE

Boston University Fall 2022, Spring 2023
Teaching Assistant, EC441: Introduction to Networking

MENTORING

Carnegie Mellon University: Fenghao Dong, Yen-Ru Chin, Ashwin Silla
University of Maryland: Eric S. Wang, Eesha Kurella
Boston University: Nengneng Yu, Rashid Kolaghassi, Maxwell Malamut, Robert D'Antonio

PROFESSIONAL SERVICE

Conference Service

Program Committee: Sensors S&P Workshop 2025

Reviewer: NeurIPS 2026, ICLR 2027

Website Chair: HotNets 2025

Shadow Program Committee: SIGMETRICS 2026, EuroSys 2026

Artifact Evaluation Committee: SIGCOMM 2025

Community Service

Event Host, N2Women Young Researcher Workshop (2022, 2026)

Cylab Student Seminar Organizer, CMU (2026)

Women in Computer Science Lunch, CMU (2025)

CRA-WP Grad Cohort for Women (2023)

REFERENCES

Prof. Zaoxing (Alan) Liu
University of Maryland, College Park
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Prof. Vyas Sekar
Carnegie Mellon University
vsekar@andrew.cmu.edu

Prof. Francis Y. Yan
University of Illinois Urbana-Champaign
fyy@illinois.edu

Dr. Kevin Hsieh
Microsoft Research
Kevin.Hsieh@microsoft.com